
YOLO11-EARTH: TRAINING A REMOTE SENSING OPEN VOCABULARY DETECTOR AT ONLY \$21

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Abstract

In the field of remote sensing images processing, conventional object detection models faces problems such as large differences in dataset label coverage and image resolution, and a lack of detection models with strong transferability. Meanwhile in computer vision community, with similar concerns, Open Vocabulary Detection (OVD) has been proposed to detect labels outside the training data. However, existing OVD models consume a large amount of accelerator memory during training and require multiple datasets from different tasks, making this solution cost high.

Aiming to bridge this gap, this technical report introduces a lightweight solution for open vocabulary detection in remote sensing images. This solution mixed YOLO11 and YOLOv8-Worldv2 and proposes a new variant, YOLO11-Earth, through model pruning. It only uses the conventional xView dataset and a consumer-grade GPU, enabling real-time open vocabulary detection capabilities on remote sensing images. In the DIOR zero-shot transfer test, YOLO11-Earth outperforms the vanilla YOLOv8-Worldv2 and is comparable to its tuned-on-xView version. In the DIOR fine-tuning test, YOLO11-Earth surpasses YOLOv8-Worldv2 and YOLO11 when using both partial and full fine-tuning training data, with an mAP50 of 84.5% for YOLO11x-Earth, demonstrating strong cross-dataset transfer ability. YOLO11-Earth can also pass the pre-trained backbone network to the YOLO11-seg model and be applied to the in-box segmentation task on the Potsdam dataset.

This study also uses machine learning theory to explain the source of YOLO11-Earth’s open vocabulary detection ability. The lightweight design of VL-PAN adds offline encoded text features as bias terms in the intermediate process of the object detection model, achieving open vocabulary capabilities but also resulting in the loss of zero-shot transferability.

Keywords: Open vocabulary detection, Remote sensing image processing, Model pruning, Transfer learning, Interpretability for AI

1 Introduction

Open Vocabulary Detection (OVD) has garnered significant attention in recent years, primarily due to its ability to identify objects or categories that a model has not encountered during its training phase [1, 2]. This capability is particularly crucial in the context of remote sensing imagery, where the diversity and complexity of objects necessitate models with robust generalization abilities.

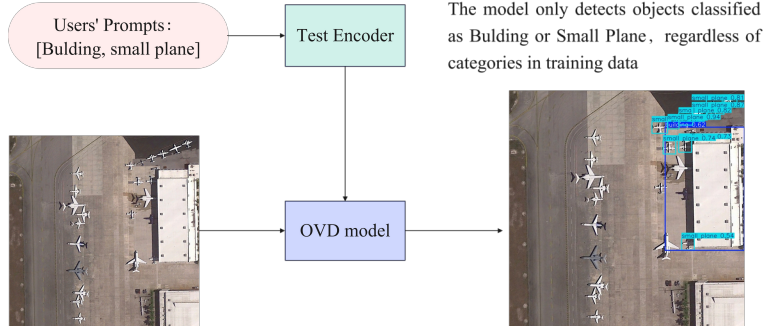


Figure 1: The "prompt first, then detect" pipeline of open vocabulary detection model.

Conventional object detection models, such as R-CNN, SSD, and YOLO, have shown remarkable performance in detecting objects and classes within their training datasets [3, 4, 5]. However, these models often struggle when deployed in new scenarios with unseen categories, as they are inherently limited by the closed-set training [6, 7, 8]. To address this limitation, OVD models have been developed to extend the detection capabilities beyond the training dataset, enabling the identification of unseen objects through the integration of additional modalities, such as textual information [9, 10, 11].

In this technical report, we develop a lightweight OVD model tailored for remote sensing imagery. Our proposed model, YOLO11-Earth, integrates the strengths of YOLO11 and YOLOv8-Worldv2 while significantly reducing the computational cost compared to SOTA (State of The Art) OVD models. This is achieved through model pruning and the use of a single, conventional dataset (xView) along with a consumer-grade GPU (RTX4090). The xView dataset, with its extensive coverage of various objects and high-resolution imagery, serves as the foundation for our pre-training process. By leveraging this dataset, we aim to equip YOLO11-Earth with the ability to perform real-time OVD on any remote sensing images, thereby overcoming the limitations of existing models that require extensive computational resources and multiple datasets for training.

The primary objectives of this research are threefold. Firstly, we focus on the design and pre-training of the lightweight OVD model, YOLO11-Earth, which involves reconstruct the Vision-Language Path Aggregation Network (VL-PAN) to enhance multimodal learning on xView. Secondly, we evaluate the cross-dataset and cross-task transfer learning capabilities of YOLO11-Earth through zero-shot transfer and fine-tuning experiments on the DIOR dataset, as well as by applying the model to the Potsdam segmentation dataset using YOLO11-seg. Lastly, we delve into the interpretability and potential improvements of YOLO11-Earth, comparing it with state-of-the-art models such as Grounding DINO and LAE-DINO to elucidate the sources of its OVD capabilities and identify areas for future enhancement.

The remainder of this paper is organized as follows. Section 3 provides a detailed description of the datasets used for pre-training and evaluation, along with the data preprocessing methods. Section 4 outlines the experimental procedures and evaluation metrics, followed by a comprehensive presentation of the YOLO11-Earth model architecture and its differences from baseline models. Section 5 presents the experimental results, including pre-training performance, cross-dataset transfer learning outcomes, and cross-task transfer learning capabilities. Finally, Section 6 analyzes the low cost advantages and interpretability of YOLO11-Earth, and discusses potential improvements and future directions.

2 Related Work

2.1 Object Detection Models

Object detection models have evolved from two-stage methods like the R-CNN series, which prioritize accuracy at a high computational cost [3, 12, 13, 14], to one-stage models such as SSD and the YOLO family, which are optimized for real-time performance [4, 5]. The YOLO series, in particular, has seen rapid development. Early versions improved network structures and multi-scale handling [15, 16]. Subsequent iterations introduced innovations like advanced loss functions [17], anchor-free mechanisms [18], module-level reparameterization [19], and attention mechanisms [20], while continuously enhancing training strategies, parameter efficiency, and deployment flexibility [21, 22, 23, 24, 25].

Concurrently, the integration of Transformer architectures has led to end-to-end detectors like DETR [26] and its successors, such as DINO, which improved performance through techniques like contrastive de-noising training [27]. Further refinements in this domain have yielded models like RT-DETR and Mask DINO, pushing performance boundaries [28, 29, 30].

2.2 Open Vocabulary Detection Models

Open Vocabulary Detection (OVD) models leverage multi-modal pre-training to align visual and textual features, enabling the detection of unseen categories. CLIP (Contrastive Language-Image Pretraining) has been a significant inspiration for OVD models, using contrastive learning to generate pseudo-labels and align text and image features [31]. Models like ViLD-text and ViLD-image have explored integrating language features into traditional object detection models like Mask-RCNN [32]. GLIP further enhanced this approach by using a Swin-Transformer backbone and optimizing the balance between classification and localization losses [33, 34, 35]. Grounding DINO combined the Transformer-based DINO detector with multi-modal pre-training, using composite datasets for language-image alignment [10]. YOLO-World introduced a real-time OVD model by integrating CLIP’s text encoder and a reparameterizable visual-language path aggregation network (RepVL-PAN) [9]. Subsequent models like YOLO-UniOW and YOLOE introduced adaptive decision learning and wildcard learning strategies to improve efficiency and adaptability [36, 37]. These models have demonstrated significant advancements in OVD, but they often require substantial computational resources for training.

2.3 Remote Sensing Image Open Vocabulary Detection

The application of OVD in remote sensing imagery has seen notable advancements with models like LAE-DINO, which introduced pseudo-label generation and text-image information filtering strategies [38]. LAE-DINO’s DVC module dynamically selects positive and negative vocabularies for each training batch, addressing the challenges of large-scale vocabulary sets. The VisGT module enhances text features by introducing “scene features,” which help align visual and textual features across the entire image. More recent models like OpenRSD have adopted multi-stage training processes—pre-training, fine-tuning, and self-training—to improve generalization capabilities [39]. These models have shown promise in enhancing the performance and efficiency of OVD in remote sensing imagery, but challenges remain in terms of computational costs and dataset requirements.

In summary, while significant progress has been made in both object detection and OVD, there is a need for lightweight models that can achieve real-time performance with reduced computational requirements. This study aims to address this gap by proposing a lightweight OVD model, YOLO11-Earth, designed specifically for remote sensing imagery.

3 Benchmark Datasets and Preprocessing

3.1 Benchmark Dataset Selection

The construction of the YOLO11-Earth model in this study follows the general schema of open vocabulary detection model construction: first conducting pre-training on the xView dataset, which has the most labels and categories. Subsequently, zero-shot transfer, few-shot fine-tuning, and full fine-tuning experiments are carried out on the DIOR dataset, which has fewer labels and categories and lower difficulty. Finally, the YOLO11-seg model is initialized with the pre-trained YOLO11-Earth backbone network and fine-tuned on

the Potsdam dataset for segmentation tasks. Unlike the majority of existing OVD research, YOLO11-Earth aims to achieve open vocabulary detection capabilities through pre-training on a closed training set.

The xView dataset [40] comprises images of complex scenes from around the world, annotated with bounding boxes and containing over a million object instances across 60 categories. The images have a spatial resolution of 0.3 m and cover an area exceeding 1,400 km² on the Earth’s surface. The dataset includes various types of land cover and land use, except for China, and retains natural noise such as clouds and fog. The label and category distribution in xView is highly imbalanced, with significant variance in the area of bounding boxes within the same category. The purpose of pre-training on this dataset is to provide the model with as much prior knowledge as possible.

The DIOR dataset [41] contains 23,463 JPEG images and 190,288 instances, covering 20 target categories with spatial resolutions ranging from 0.5 m to 30 m. Similar to xView, the imaging conditions, weather, seasons, and image quality vary greatly, resulting in high inter-class similarity and intra-class diversity. This benchmark helps assess the cross-dataset transfer learning capabilities of YOLO11-Earth after pre-training. The dataset is widely used for evaluating remote sensing object detection models. In this study, following the LAE-DINO research, the training set of the DIOR dataset is further sampled to simulate the model’s transfer learning capabilities under data scarcity conditions.

The Potsdam segmentation dataset [42], provided by the International Society for Photogrammetry and Remote Sensing (ISPRS), consists of 38 orthoimages with a resolution of 5 cm, covering a typical historical urban area (the city of Potsdam). Each image is annotated with detailed pixel-level labels for six main categories: background, car, tree, low vegetation, building, and impervious surfaces. These annotations provide reliable ground truth data for algorithm training and performance evaluation. The dataset’s high resolution and scene diversity allow for the clear depiction of land cover details and encompass various scenes, including urban, suburban, and natural landscapes. The dataset’s moderate size (equivalent to a medium-sized European city) makes it suitable for fine-tuning in cross-task scenarios.

3.2 Data Preprocessing

The xView dataset was manually annotated using QGIS, with the original labels in GeoJSON format recording the category and absolute position of the bounding boxes within the images. The images in the xView dataset have varying sizes and are stored in TIFF format. During experiments, it was found that these characteristics made it difficult to fully utilize the limited 24 GB of GPU memory, and memory fluctuations easily caused memory leaks, leading to training interruptions. First, according to the GeoJSON labels, all annotations were divided into TXT files by image name, and the coordinates were normalized to the image range to obtain YOLO-format labels. To address the issue of low GPU memory utilization efficiency and the negative correlation with model receptive field size, the OpenCV library was used to losslessly convert the TIFF images into PNG images with significantly reduced storage space. Additionally, to solve the problem of small-scale information loss due to the model’s receptive field being much smaller than the input image size, we adopted the slicing-assisted fine-tuning (SAFT) idea proposed for the DOTA dataset. The original large and inconsistent-sized remote sensing images were divided into 1024×1024 pixels slices using a sliding window method, generating overlapping sub-images so that targets would not be lost at slice boundaries (Figure 2). We rewrote the DOTA devkit segmentation script using the Shapely library and the same IoU calculation and label-matching algorithm, discarding labels with $\text{IoU} < 0.7$. With an overlap rate of 0.2 (200 pixels), we obtained 9,442 sub-images and corresponding label files, then split them into training and validation sets at an 8:2 ratio. The validation set contains 1,889 images with 203,917 instances.

The DIOR dataset uses JPEG format with lossy compression, with each image fixed at 800×800 pixels, saving a significant amount of local storage. The label files are in xml format (Pascal VOC standard format). To better train a YOLO series derived model, the original xml dataset was converted to txt text files, and the coordinates were normalized. The dataset split followed the DIOR paper, with 11,725 images as the train/val dataset and 11,738 images as an independent test set to evaluate model performance after training. The train/val dataset was split at an 8:2 ratio. Additionally, to verify the few-shot transfer capabilities of the open vocabulary detection model, the training set was further sampled independently at 25% and 50% proportions to obtain the train-quarter and train-half datasets. This step aims to simulate fine-tuning tasks under data scarcity conditions.

The Potsdam segmentation dataset preprocessing involved converting GeoTIFFs to JPEG, tiling into 224×224 pixel sub-images without overlap, and mapping mask pixels to YOLO-format instance-level segmentation TXT label files (COCO-seg style). Figure 3 illustrates this process.

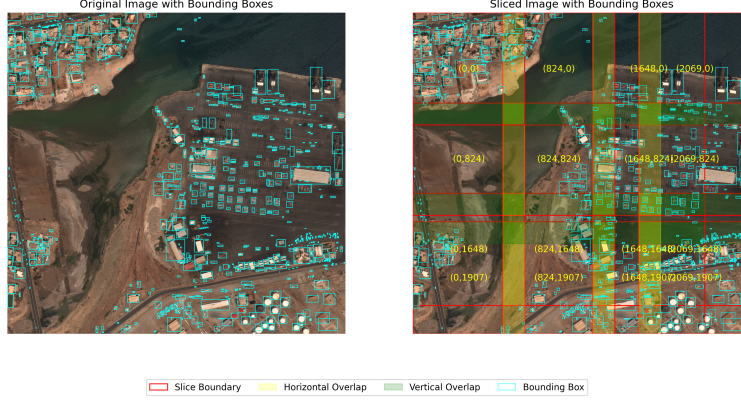


Figure 2: Visualization of the data slicing process for the Potsdam dataset. The original high-resolution remote sensing images are divided into 224×224 pixel sub-images to facilitate training while preserving the detailed information within each tile.

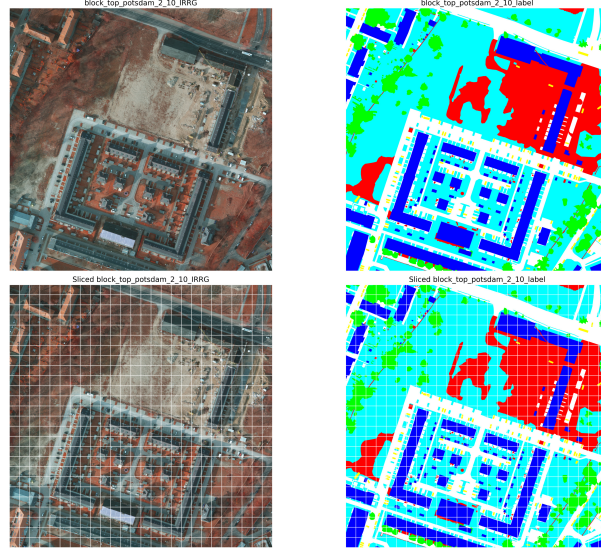


Figure 3: Visualization of the data slicing process for the Potsdam dataset. The original high-resolution remote sensing images are divided into 224×224 pixel sub-images to facilitate training while preserving the detailed information within each tile.

4 Construction of Lightweight Remote Sensing Open Vocabulary Detector

4.1 Experimental Workflow and Evaluation Metrics

In the context of datasets, traditional supervised learning datasets have a fixed encoding of supervision signals for each input element. For example, an image labeled as “cat” belongs to a closed set, where the image labels have a predefined range, and traditional models can only map predictions to one of the predefined categories through probability.

Before the advent of open vocabulary detection (OVD) tasks, contrastive training methods in language-image contrastive learning allowed models to generate pseudo-labels for each image instance, resulting in an open set where the training dataset is not hard-encoded and can be freely defined after training.

Existing OVD detectors and open-set detectors follow the pre-training strategy of language-image contrastive learning and the visual grounding task in multi-modal large models, converting multiple closed-set training datasets online into an open set. For instance, YOLO-World employs a large-scale object detection dataset, two localization datasets, and a language-image contrastive learning dataset for multi-modal pre-training.

This pre-training approach demands substantial computational resources for datasets and hardware, far exceeding the capabilities of average users. Therefore, we attempt to conduct multi-modal pre-training on a much smaller closed-set benchmark dataset, xView, to achieve zero-shot transfer capabilities across datasets. This capability in OVD models is manifested as the ability to recognize new targets by simply adjusting the category names (prompt fine-tuning) without retraining.

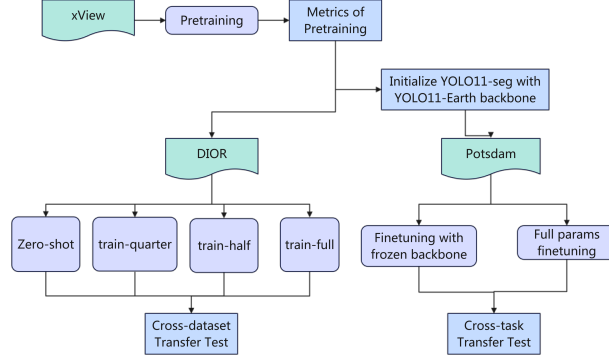


Figure 4: Experimental workflow. The model is first pre-trained on xView, then evaluated cross-dataset transfer ability through zero-shot transfer and fine-tuning on DIOR, and finally tested for cross-task transfer on Potsdam.

The experimental framework of this study is illustrated in Figure 4. We first conduct pre-training on the traditional dataset xView and compare the precision of different models on the pre-training dataset. Subsequently, we use two benchmark datasets, DIOR and Potsdam, to evaluate whether the pre-training has endowed the model with OVD capabilities and its generalization capabilities across different tasks (object detection and instance segmentation).

Specifically, on the DIOR dataset, we perform zero-shot transfer, 25% training data fine-tuning, 50% training data fine-tuning, and full training data fine-tuning experiments to assess detection accuracy and evaluate the cross-dataset transfer learning capabilities of YOLO11-Earth. On the instance segmentation dataset Potsdam, we initialize the YOLO11-seg segmentation model with the pre-trained YOLO11-Earth backbone network and conduct fine-tuning with and without freezing the backbone network to evaluate segmentation accuracy and assess the cross-task transfer learning capabilities of YOLO11-Earth.

In object detection and instance segmentation tasks, we use a series of model evaluation metrics to measure model performance. These metrics include precision (P), recall (R), mean average precision at an IoU threshold of 0.5 (mAP50), and mean average precision across IoU thresholds from 0.5 to 0.95 (mAP50:95). Precision measures the accuracy of the model’s positive predictions, while recall measures the model’s ability to identify all true positives. The formulas for precision and recall are as follows:

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

where TP (True Positives) represents correctly identified positive instances, FP (False Positives) represents incorrectly identified positive instances, and FN (False Negatives) represents missed positive instances. mAP50 and mAP50:95 are more comprehensive evaluation metrics, representing the mean average precision at an IoU threshold of 0.5 and across IoU thresholds from 0.5 to 0.95, respectively. These metrics provide a more holistic assessment of model performance under different conditions. The calculation formula for mAP50:95, using a discrete summation method with a step size of 0.05, is as follows:

$$mAP_{50:95} = \frac{1}{10} \sum_{t=0.5}^{0.95} AP_t \quad (3)$$

where AP_t is the average precision at IoU threshold t , calculated as the area under the precision–recall curve. The mean average precision (mAP) is calculated as follows: is the average precision of the i -th category.

In the segmentation task on the Potsdam dataset, we use the same metrics for mask evaluation, calculated in the same manner as for object detection tasks.

In addition to precision performance metrics, we also use GFLOPS (Giga Floating Point Operations Per Second) and Params (model parameter count) to measure model computational cost. GFLOPS indicates the number of floating-point operations per second, reflecting model computational complexity, with its calculation involving model parameter count and input image size. Params represents the total number of trainable parameters in the model, calculated as the product of the number of neurons in each layer. These two parameters are positively correlated with model computational cost.

Through these metrics, we can comprehensively evaluate the performance and efficiency of YOLO11-Earth as a real-time object detection and instance segmentation model and compare it with existing research.

Since the outstanding LAE-DINO in related works did not provide the specific DIOR dataset split used in their paper and the model weights were not open-sourced at the time of our experiments, we could not conduct strict comparative experiments. Instead, we used the data from their paper for comparison while listing the metrics of YOLO11-Earth on the DIOR-val and DIOR-test datasets to rigorously compare model performance.

4.2 Model Design and Implementation

4.2.1 Model Architecture

We redesigned the VL-PAN to enable YOLO11 to learn multi-modal priors. Based on this structure as the neck of YOLO11-Earth and using the multi-modal detection head (WorldDetect), we introduced the backbone network and network depth strategy of the YOLO11 series to replace the YOLOv8 detector in YOLO-Worldv2. The complete model architecture and forward propagation process of YOLO11-Earth are shown in Figure 4, where: – conv denotes a plain two-dimensional convolutional layer, – k represents the kernel size, – s represents the stride. The YOLO11 model family’s greatest contribution lies in the use of the

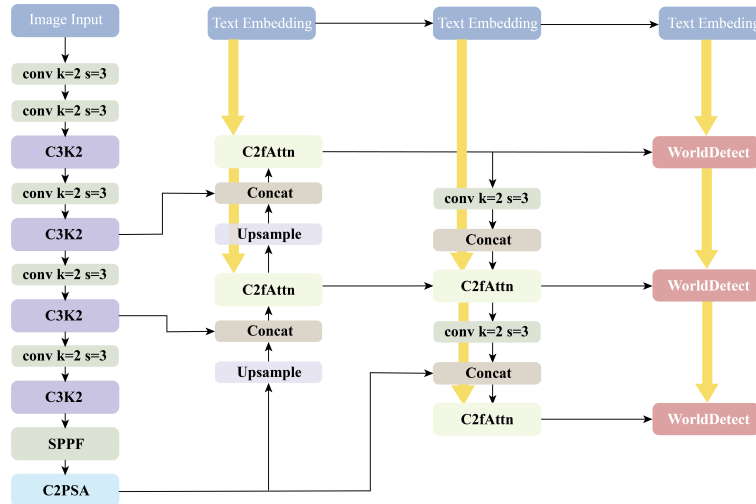


Figure 5: The architecture of YOLO11-Earth. The model consists of a CSPNet backbone, a Vision-Language Path Aggregation Network (VL-PAN) for feature fusion, and a WorldDetect head for multi-modal detection. Text features from the CLIP text encoder are integrated at multiple levels of the network.

C3K2 module to achieve greater depth while maintaining gradient flow and fewer parameters compared to YOLOv8 (and even slightly better precision and speed on COCO compared to YOLOv9 and YOLOv10).

In simple terms, YOLO11-Earth can be seen as either: 1.A YOLO-Worldv2 model updated with a new backbone network and trained on a small closed-set multi-modal scheme, 2.A YOLO11 model endowed with new multi-modal learning capabilities, retrained on remote-sensing benchmarks and adapted to application scenarios very different from YOLO-Worldv2.

This study uses the Ultralytics PyTorch implementation of YOLO-Worldv2 (YOLOv8-Worldv2), which is easier to deploy and train. Apart from the non-reparameterizable VL-PAN part, it is identical to the mncv-based YOLO-Worldv2 and shares the same pretrained weights.

We denote model scales by appending s, m, l, or x to the version number (e.g., YOLO11s-Earth, YOLOv8x-Worldv2).

Figure 5 shows the detailed internal parameter layers of YOLO11-Earth.

4.2.2 VL-PAN Design for Close-Set Datasets

The original YOLO-World architecture includes a language-image alignment module that cannot be deployed on consumer-grade GPUs. Although this module can maximize the interaction between text and image information, it consumes a significant amount of computational resources. Therefore, we prune this part.

Meanwhile, since the labels in xView are hard-encoded (fixed text descriptions), we cannot train text embeddings or classification heads. Removing this part and retaining text features as inputs to the model’s neck is a natural choice.

Thus, we adjust the RepVL-PAN (Reparameterizable Vision-Language Path Aggregation Network) in YOLO-Worldv2, retaining the T-CSP block (mainly composed of C2FAttn modules) as part of the YOLO model’s neck and removing the original text embedding component for language-image contrastive learning to complete the pruning. As shown in Figure 6, after aggregating text encoding into image features, YOLO11-Earth does not update the text encoding.

Finally, we add a step to feed text encoding into the detection head, enabling it to perceive information from both modalities and improve cross-dataset transfer capabilities.

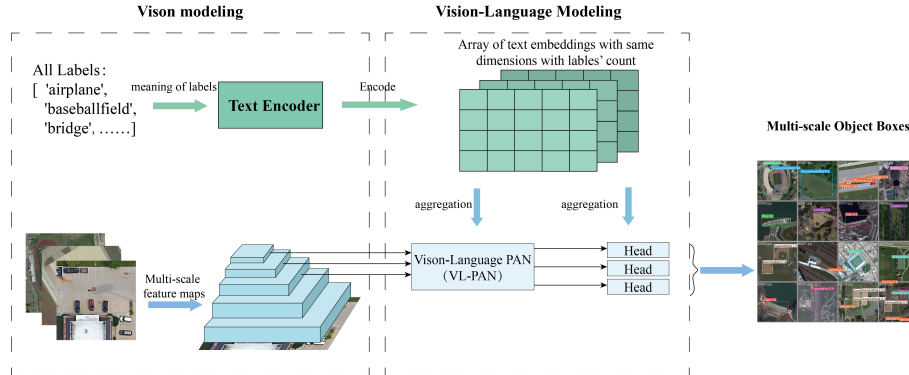


Figure 6: Multi-Modal workflow of YOLO11-Earth. The model aggregate the text embeddings into the image features at multiple levels. The text features are encoded once and passed through the network as a feature supplement to the traditional YOLO11 model.

Overall, the redesigned VL-PAN (Vision-Language Path Aggregation Network) no longer updates and forwards text encoding or calculates losses. It only encodes all input label texts once and passes them through the network as a feature supplement to the traditional YOLO model. This text feature can be encoded once by CLIP (the chosen text encoder) during each training iteration and stored, significantly reducing the model’s actual computational complexity. Additionally, we no longer require a special text embedding loss function but can directly apply the YOLO11 training loss, which consists of three parts: bounding box loss, classification loss, and DFL (Distribution Focal Loss).

It is worth noting that since the semantic alignment module in YOLO-Worldv2 does not participate in the forward process during inference (offline vocabulary mechanism), we can still load its pre-trained weights published in the paper for ablation experiments on the DIOR test set.

In summary, the VL-PAN design in YOLO11-Earth effectively integrates text features into the image feature extraction process, enhancing the model’s ability to generalize across different datasets while maintaining computational efficiency. This design choice allows YOLO11-Earth to leverage the benefits of multi-modal learning without the prohibitive computational costs associated with more complex language-image alignment modules.

5 Experimental Results and Analysis

5.1 Pretraining on Closed-set Datasets

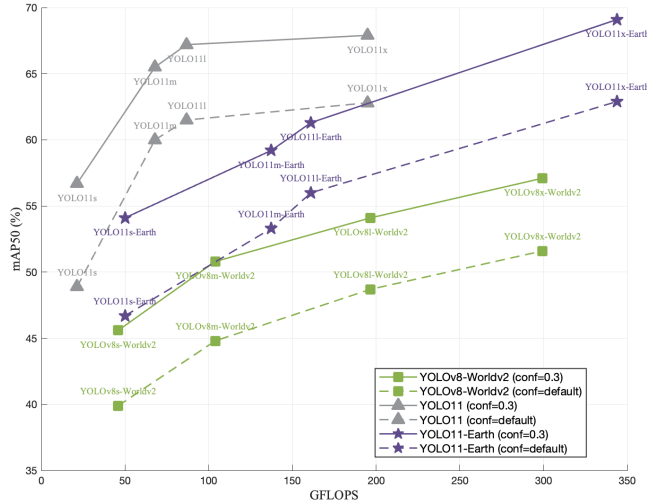


Figure 7: Pretraining results of YOLO11-Earth and baseline models on xView-valid.

Table 1 presents the evaluation results of the 12 models across three model families on the validation set. On the xView validation set, we found that the pre-trained YOLO11-Earth significantly outperformed the YOLOv8-Worldv2 model trained on the same dataset and nearly matched the closed-set object detector YOLO11, which was also fully trained. During the precision assessment of the pre-trained models, we observed that the YOLO-World architecture (YOLOv8-Worldv2 and YOLO11-Earth) tended to detect more false positives (small targets with low confidence) in the validation set. However, by setting a lower confidence threshold, the precision of bounding boxes could be significantly improved, indicating that our models might overly focus on fine-grained features, leading to incorrect target detection in the background. Figure 6 more intuitively displays the performance differences among the three model families shown in Table 1. The use of a $\text{conf}=0.3$ filtering operation improved the mAP50 metric for all 12 models. Therefore, in subsequent experiments (including DIOR), we set the confidence threshold for evaluation tasks to $\text{conf}=0.3$. In addition to the performance boost brought by $\text{conf}=0.3$, Figure 6 also shows that YOLO11, as a single-modal model, exhibits lower inference costs.

Compared with YOLO11-Earth and YOLOv8-Worldv2, this performance difference implies that the improved backbone network has led to the performance gains of YOLO11-Earth. Directly using remote sensing images for pretraining offers the advantage of better approximating the model’s global optimal parameters. Especially when comparing YOLOv8l-worldv2 with YOLO11m-worldv2, the latter, with 40% fewer parameters, significantly outperforms the former, which roughly reflects the original performance difference between the YOLOv8 and YOLO11 series. This indicates that YOLO11-Earth has effectively inherited the feature extraction advantages of YOLO11. Compared with the closed-set detector YOLO11, YOLO11-Earth is significantly weaker in the s, m, and l sizes. This gap suggests that although the C2FAttn module acquires multimodal learning capabilities through attention mechanisms, it sacrifices a considerable portion of the pure image feature fitting capability compared to the C3K2 module. However, YOLO11x-Earth, by employing a larger parameter count, not only matches the object detection capability of YOLO11x but also gains the multimodal fusion capability absent in the closed-set detector (Table 1). This implies that the open vocabulary detection architecture typically requires a larger parameter count to balance the computation tasks of semantic and image features. As the parameter count increases, the model’s ability to fit complex data distributions may improve. To test whether our block strategy, in addition to reducing GPU runtime pressure, can improve the model’s ability to perceive small targets in large-size images, we evaluated the block-pretrained models on the original-size xView validation set. The performance metrics are shown in Table 2. The results indicate that even though the models were trained on block images, all our open vocabulary detection models could still perform efficient 1024×1024 inference on the original validation

Table 1: Metrics of YOLO11-Earth and baseline models on xView. In parentheses is the change of indicator relative to conf=default after conf=0.3 is used.

Model Structure	P(%)	R(%)	mAP50(%)	mAP50:95(%)	GFLOPS
YOLOv8s-Worldv2	52.1 (+0.5)	36.5 (-6.5)	45.6 (+5.7)	29.2 (+5.4)	46
YOLOv8m-worldv2	55.9 (-0.9)	42.1 (+0.22)	50.8 (+6)	33.3 (+5.8)	103.9
YOLOv8l-Worldv2	57.9 (-0.5)	46.1 (-1.7)	54.1 (+5.4)	36.3 (+5.7)	196.6
YOLOv8x-Worldv2	59.4 (-1.2)	49.7 (+0)	57.1 (+5.5)	38.7 (+5.7)	299.4
YOLO11s-Earth	65.4 (+7.1)	40.4 (-5.9)	54.1 (+7.4)	36.2 (+7.2)	50.0
YOLO11m-Earth	64.2 (-1.0)	49.4 (-1.0)	59.2 (+5.9)	41.0 (+6.2)	137.4
YOLO11l-Earth	63.6 (-0.4)	53.2 (-0.8)	61.3 (+5.3)	43.0 (+5.7)	161.1
YOLO11x-Earth	73.4 (+1.8)	58.1 (-0.9)	69.1 (+6.2)	51.8 (+7.2)	344.0
YOLO11s	68.4 (+6.2)	42.5 (-5.5)	56.7 (+7.8)	38.1 (+7.6)	21.4
YOLO11m	71.8 (+0)	54.5 (-0.6)	65.5 (+5.5)	46.4 (+6.3)	67.9
YOLO11l	71.2 (+0.3)	57.9 (-0.1)	67.2 (+5.7)	47.7 (+6.5)	86.8
YOLO11x	72.2 (-1.8)	58.3 (+0.1)	67.9 (+5.1)	49.0 (+6.0)	194.8

set and achieve performance comparable to state-of-the-art models. However, since our sub-images might contain partial information from the original validation set and the mAP50 metric is relatively more lenient than the mAP metric, we cannot conclude that our models have surpassed the MTP model’s performance [49]. Nevertheless, this performance demonstrates that training on block images can enhance the model’s inference effects on larger input images. This improvement originates from the model parameters’ fitting to finer-grained features and the repeated learning of difficult targets due to overlapping sub-images. This also corroborates the effectiveness of the SF method and the corresponding SAHI method for small target detection in very large images. Although our training and inference did not directly use the SAHI library, the principles are essentially the same. Even though the models were trained on block images, all our open

Table 2: Metrics of YOLO11-Earth and baseline models on the original-size xView validation set.

Model Structure	P	R	mAP50	mAP50:95	GFLOPS
YOLOv8s-Worldv2	21.0%	3.75%	12.3%	7.40%	46
YOLOv8m-Worldv2	26.1%	6.20%	15.9%	8.73%	103.9
YOLOv8l-Worldv2	27.4%	6.87%	17.1%	9.89%	196.6
YOLOv8x-Worldv2	35.2%	7.60%	21.1%	11.90%	299.4
YOLO11s-Earth	26.7%	6.15%	16.1%	9.85%	50.0
YOLO11m-Earth	32.6%	7.73%	20.3%	11.60%	137.4
YOLO11l-Earth	34.7%	8.73%	21.5%	11.60%	161.1
YOLO11x-Earth	36.7%	9.31%	22.9%	13.90%	344.0

vocabulary detection models could still perform efficient 1024×1024 inference on the original validation set and achieve performance comparable to state-of-the-art models. However, since our sub-images might contain partial information from the original validation set and the mAP50 metric is relatively more lenient than the mAP metric, we cannot conclude that our models have surpassed the MTP model’s performance [43]. Nevertheless, this performance demonstrates that training on block images can enhance the model’s inference effects on larger input images. This improvement originates from the model parameters’ fitting to finer-grained features and the repeated learning of difficult targets due to overlapping sub-images. This also corroborates the effectiveness of the SF method and the corresponding SAHI method for small target detection in very large images. Although our training and inference did not directly use the SAHI library, the principles are essentially the same.

5.2 Cross-dataset Transfer Results

5.2.1 Zero-Shot Transfer Results

Table 3 shows the performance comparison of YOLO11-Earth and baseline models on the DIOR test set for zero-shot transfer. To evaluate the zero-shot transfer capabilities of our open vocabulary detector on an

unseen dataset, we used the 12 pre-trained models from Section 5.1, as well as the pre-trained weights of YOLOv8-Worldv2 from its paper, and conducted inference and evaluation on the DIOR test set by merely modifying the vocabulary.

Table 3: Metrics of YOLO11-Earth and baseline models on the DIOR test set for zero-shot transfer.

Model Structure	P	R	mAP50	mAP50:95	GFLOPS
YOLOv8s-Worldv2	7.74%	11.5%	8.12%	3.67%	46
YOLOv8m-Worldv2	7.87%	11.9%	8.11%	2.98%	103.9
YOLOv8l-Worldv2	7.87%	11.9%	8.11%	2.98%	196.6
YOLOv8x-Worldv2	10.40%	15.6%	9.95%	3.57%	299.4
YOLOv8s-Worldv2 (untrained on xView)	10.30%	2.62%	2.10%	1.06%	51.0
YOLOv8m-Worldv2 (untrained on xView)	38.20%	2.10%	3.27%	1.85%	110.5
YOLOv8l-Worldv2 (untrained on xView)	27.90%	3.03%	3.49%	1.92%	204.5
YOLOv8x-Worldv2 (untrained on xView)	14.60%	4.64%	4.49%	2.51%	309.3
YOLO11s-Earth	8.20%	6.09%	6.26%	2.98%	50.0
YOLO11m-Earth	10.40%	7.72%	7.79%	3.92%	137.4
YOLO11l-Earth	10.20%	7.23%	7.15%	3.47%	161.1
YOLO11x-Earth	12.90%	3.92%	7.95%	4.35%	344.0
YOLO11s	-	-	-	-	21.4
YOLO11m	-	-	-	-	67.9
YOLO11l	-	-	-	-	86.8
YOLO11x	-	-	-	-	194.8

The results indicate that pretraining on the xView dataset significantly enhances the zero-shot transfer capabilities of the open vocabulary detection models on remote sensing images. Comparing the YOLOv8-Worldv2 models trained with and without xView, it is evident that including remote sensing images in the training set greatly improves the model’s ability to generalize to new datasets. However, YOLO11-Earth, despite its superior performance during pretraining, exhibits slightly weaker zero-shot capabilities on the DIOR dataset compared to YOLOv8-Worldv2. This discrepancy may arise from the closed-set training data’s inability to enforce alignment between language and image features, leading to an overemphasis on image features. Meanwhile, the closed-set single-modal detector YOLO11 completely lacks open vocabulary detection capabilities. For instance, YOLO11s achieved an mAP50 of 58.9% on the first label, 0.66% on the 14th label, and 0.45% on the 19th label, with zero precision for the remaining labels. This inconsistent performance demonstrates that YOLO11 cannot reliably classify objects in the DIOR dataset, as the labels it learned from xView do not match those in DIOR.

5.2.2 Full Fine-tuning Transfer Results

After the zero-shot transfer experiments, we conducted fine-tuning experiments using the complete training set (DIOR-train) for all 12 models. In this scenario, the open vocabulary detectors and the traditional closed-set detectors perform the same task, and we aim to verify whether the open vocabulary detectors with semantic information can achieve learning with fewer training iterations (80–120 iterations with early stopping).

Table 4 shows the performance metrics of the pre-trained models on the validation and test sets after fine-tuning on the DIOR dataset. Figure 7 provides a more intuitive comparison of the models’ performance.

The results demonstrate that YOLO11-Earth, based on YOLO11 and YOLOv8-Worldv2, is a successful practice of promoting closed-set detectors to open vocabulary detection. An important finding is that after fine-tuning, the computational cost (GFLOPS) of the two open vocabulary models dropped significantly, while the parameter reduction in YOLO11 was minimal. This difference is likely due to the DIOR dataset having shorter labels and 40 fewer categories compared to the xView dataset. In the two open vocabulary detection models, labels are stored as text features, whereas in YOLO11, they correspond one-to-one with the fully connected layer’s numerical mapping, as in most single-modal models.

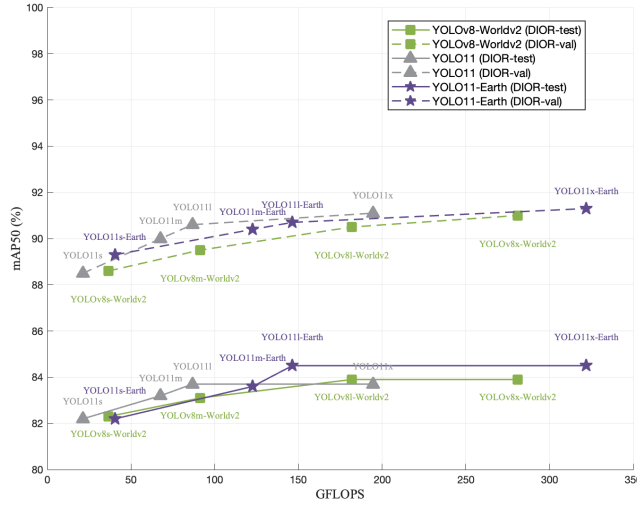


Figure 8: Fine-tuning with full train set results of YOLO11-Earth and baseline models on DIOR.

Table 4: Metrics of YOLO11-Earth and baseline models on the DIOR validation and test sets after full fine-tuning.

Model Structure	mAP50-val	mAP50:95-val	mAP50-test	mAP50:95-test	GFLOPS
YOLOv8s-Worldv2	88.6%	69.4%	82.3%	64.3%	36.6
YOLOv8m-Worldv2	89.5%	72.3%	83.1%	66.2%	91.3
YOLOv8l-Worldv2	90.5%	73.5%	83.9%	67.0%	181.9
YOLOv8x-Worldv2	91.0%	74.0%	83.9%	67.3%	281.1
YOLO11s-Earth	89.3%	71.1%	82.2%	64.8%	50.0
YOLO11m-Earth	90.4%	71.6%	83.6%	65.5%	122.7
YOLO11l-Earth	90.7%	72.7%	84.5%	66.8%	161.1
YOLO11x-Earth	91.3%	74.2%	84.5%	67.2%	322.0
YOLO11s	85.5%	69.6%	82.2%	64.4%	21.3
YOLO11m	90.0%	72.2%	83.2%	65.7%	67.7
YOLO11l	90.6%	73.9%	83.7%	66.9%	86.7
YOLO11x	91.1%	74.3%	83.7%	67.1%	194.5

5.2.3 Less Data Fine-tuning Transfer Results

Similar to LAE-DINO, to investigate the transfer learning capabilities of YOLO11-Earth under data scarcity conditions, we conducted fine-tuning experiments on the train-quarter and train-half datasets, which were randomly sampled from the full DIOR training set.

YOLO11 exhibited the weakest performance among the three model families. Even with rich pre-trained weights, it struggled to converge quickly under data scarcity conditions. This indicates that single-modal models are more prone to falling into local optima when forced to converge rapidly compared to open vocabulary detectors. As expected, the multi-modal information learned by YOLO-Worldv2 and YOLO11-Earth facilitated quicker training and superior performance. However, there were differences in the ease of parameter convergence among the models. To ensure the significance of the ablation experiments, we did not repeatedly adjust the hyperparameters and used the same default settings for all models of the same size. Compared to YOLOv8-Worldv2, the two YOLO11-based networks were more prone to early stopping on the 25% training dataset, resulting in slightly lower metrics for YOLO11-Earth. Nevertheless, YOLO11-Earth was relatively easier to train. The models’ depth (number of parameter layers) was in the order of YOLO11-Earth, YOLO-Worldv2, and YOLO11, suggesting that training difficulty is proportional to the number of parameter layers, and additional modal information helps mitigate this difficulty. When the training dataset size reached 50%, YOLO11-Earth’s performance advantage became more pronounced. The

Table 5: Comparison of YOLO11-Earth with state-of-the-art models on the DIOR test set.

Model Structure	Backbone Network	Pretraining Data	mAP50-test (%)
<i>General Object Detection Models</i>			
GASSL [44]	ResNet-50	-	67.40
CACO [45]	ResNet-50	Sentinel-2	66.91
TOV [46]	ResNet-50	TOV-NI, TOV-R	70.16
Scale-MAE [47]	ViT-L	FMoW	73.81
SatLas [48]	Swin-B	SatlasPretrain	74.10
RingMo [49]	Swin-B	RingMoPretrain	75.90
SkySense [50]	Swin-H	Multi-modal RSI	78.73
MTP [43]	Swin-H	MillionAID	81.10
<i>Open Vocabulary (Open Set) Detection Models</i>			
GLIP-FT [33, 38]	Swin-T	O365, GoldG, CC3M, SBU	87.8
GroundingDINO-FT [10, 38]	Swin-T	O365, GoldG, Cap4M	90.4
GroundingDINO-FT [10, 38]	Swin-T	LAE-1M	91.1
LAE-DINO-FT [38]	Swin-T	O365, GoldG, Cap4M	92.0
LAE-DINO-FT [38]	Swin-T	LAE-1M	92.2
YOLO11x-Earth (ours)	CSPNet	xView	84.5

two multi-modal models were relatively easier to train, while YOLO11 still faced challenges in converging quickly and often terminated prematurely. At this dataset size, we observed that the l and x-sized models performed similarly on the DIOR test set, indicating that the 50% DIOR training dataset was insufficient to support the x-sized model with over 70M parameters. Interestingly, larger models with more parameters did not necessarily outperform smaller models, as seen in the comparison between YOLO11x and YOLO11l. This suggests that optimizing a more complex non-linear mapping is challenging with a reduced training dataset size. This conclusion is supported by the performance on the larger and more challenging xView dataset, where we have demonstrated the scaling laws within each model family.

Table 6 provides a comprehensive comparison of the mAP50 dynamics for the models fine-tuned on different-sized training datasets. YOLO11-Earth consistently outperformed the other models on the test set across all three training scenarios (25%, 50%, and full training data), and matched YOLOv8-Worldv2 on the validation set. YOLOv8-Worldv2 showed good performance on the validation set but lagged on the test set. Meanwhile, YOLO11 struggled to leverage its architectural advantages under data scarcity and strict iteration limits, and could not compete with YOLO11-Earth and YOLOv8-Worldv2 even with advanced neck and head designs. This dynamic change highlights the contribution of the multi-modal input in open vocabulary detectors to the model’s ability to quickly adapt to new datasets. Clearly, YOLO11-Earth inherited the cross-dataset rapid transfer capabilities similar to the YOLO-World architecture, especially when compared to YOLO11 under data scarcity conditions. This confirms that the pretraining phase of our lightweight OVD model indeed endows YOLO11-Earth with the corresponding capabilities.

5.3 Cross-task Transfer Results

To evaluate the cross-task transfer learning capabilities of YOLO11-Earth, we conducted experiments using the YOLO11-seg model initialized with the pre-trained YOLO11-Earth backbone network on the Potsdam segmentation dataset. Specifically, we fine-tuned the YOLO11-seg model with and without freezing the backbone network to assess the segmentation performance on the Potsdam dataset.

Table 7 presents the performance metrics of the YOLO11-seg model fine-tuned with the YOLO11-Earth weights on the Potsdam dataset. The results indicate that the pre-trained backbone network of YOLO11-Earth significantly enhanced the convergence speed of the YOLO11-seg model on the Potsdam dataset, regardless of whether the backbone network was frozen or not. When the backbone network was frozen, the model achieved reasonable segmentation performance with fewer iterations, demonstrating the effectiveness of the pre-trained weights as a strong prior for the segmentation task. When the backbone network was unfrozen, allowing the weights to be updated during fine-tuning, the segmentation performance further improved, indicating that the pre-trained backbone network could adapt to the new task and dataset.

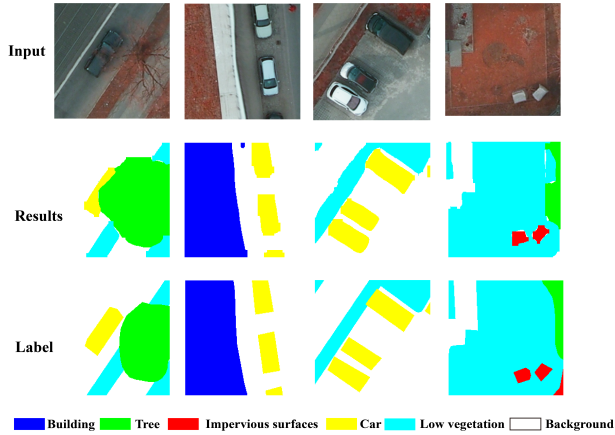
Table 6: Model performance comparison with different training data sizes.

Model Structure	Train-Quarter		Train-Half		Train-Full	
	mAP50-val	mAP50-test	mAP50-val	mAP50-test	mAP50-val	mAP50-test
YOLOv8s-Worldv2	82.0%	75.0%	84.5%	77.9%	88.6%	82.3%
YOLOv8m-Worldv2	85.0%	77.5%	85.3%	78.6%	89.5%	83.1%
YOLOv8l-Worldv2	85.0%	77.6%	88.2%	81.1%	90.5%	83.9%
YOLOv8x-Worldv2	85.6%	77.7%	88.6%	81.0%	91.0%	83.9%
YOLO11s-Earth	82.5%	75.6%	86.3%	78.8%	89.3%	82.2%
YOLO11m-Earth	84.0%	76.7%	87.6%	80.7%	90.4%	83.6%
YOLO11l-Earth	84.3%	77.3%	88.0%	81.1%	90.7%	84.5%
YOLO11x-Earth	84.4%	77.7%	88.5%	81.3%	91.3%	84.5%
YOLO11s	78.2%	70.5%	84.0%	76.3%	85.5%	82.2%
YOLO11m	81.2%	73.2%	87.1%	79.6%	90.0%	83.2%
YOLO11l	83.8%	75.9%	88.4%	80.6%	90.6%	83.7%
YOLO11x	83.2%	75.5%	88.3%	80.0%	91.1%	83.7%

Table 7: Performance metrics of the YOLO11-seg model on the Potsdam dataset.

Model	P	R	mAP50	mAP50:95	Best at(epochs)
YOLO11s-Earth (Frozen)	68.9%	56.1%	65.7%	45.1%	42
YOLO11m-Earth (Frozen)	71.8%	59.0%	68.3%	47.6%	58
YOLO11l-Earth (Frozen)	73.4%	59.5%	69.0%	48.8%	52
YOLO11x-Earth (Frozen)	72.3%	61.1%	69.4%	49.9%	49
YOLO11s-Earth (Unfrozen)	72.0%	62.1%	70.2%	49.6%	60
YOLO11m-Earth (Unfrozen)	74.2%	63.8%	71.9%	51.6%	48
YOLO11l-Earth (Unfrozen)	75.7%	63.8%	72.3%	53.0%	55
YOLO11x-Earth (Unfrozen)	76.1%	63.7%	72.7%	53.8%	38

Figure 9 illustrates the segmentation results of the YOLO11x-seg models with YOLO11-Earth backbone on the test set of the Potsdam dataset. The segmentation results were visualized by converting the bounding box predictions back to masks using RGB mapping. The model showed good performance in segmenting objects with relatively fixed shapes, such as trees and cars. However, for objects with more diverse shapes, such as impervious surfaces, sparse vegetation, and buildings, the segmentation accuracy was lower. This is partly due to the design of the Potsdam dataset, which includes a background class that leads to significant overlap of bounding boxes, making it challenging to achieve accurate pixel-level instance segmentation. In conclusion, the

**Figure 9:** Visualization of the segmentation results of the YOLO11x-Earth model on the Potsdam dataset.

pre-trained YOLO11-Earth backbone network demonstrated strong cross-task transfer learning capabilities,

enabling the YOLO11-seg model to achieve rapid convergence and reasonable segmentation performance on the Potsdam dataset. This experiment confirmed the generalization ability of the YOLO11-Earth model across different tasks, further validating its potential as a lightweight open vocabulary detection model for remote sensing imagery.

6 Cost-effectiveness and Interpretability

6.1 Cutting Down the Training Cost

Currently, SOTA models in the field of open vocabulary detection and open set detection, such as Grounding DINO and its 1.5 version, have adopted a training strategy similar to CLIP’s text-image contrastive learning. Grounding DINO integrates a Transformer-based DINO detector with multi-modal pretraining, using composite datasets (detection, localization, classification, etc.) to align language and image content, thereby enabling the detection of arbitrary targets based on category names or descriptive expressions provided by humans. However, this approach inevitably incurs high training and inference costs. Grounding DINO requires 64 Nvidia A100 GPUs to achieve a batch size of 64, and its 1.5 version, which aims to be more edge-friendly, still only achieves around 20 FPS on an A100 GPU without the acceleration of the TensorRT framework (compared to YOLO-Worldv2-L’s inference speed of 37.4 FPS).

In contrast, the first real-time open vocabulary detection model, YOLO-World, and its v2 version almost entirely emulate Grounding DINO’s training strategy. The key difference lies in the direct modification of the real-time object detection model YOLOv8, incorporating text information into the Rep-VL-PAN (corresponding to the neck of the original detector) and fusing image features with text information before performing text classification. This approach retains YOLOv8’s real-time detection capabilities, with text feature classification activated only during training. Due to the simple model structure and fewer training datasets, YOLO-World reduces the training cost of open set target detection to 32 NVIDIA V100 GPUs, achieving a batch size of 512.

The best current remote sensing open vocabulary detection model, LAE-DINO, further reduces the cost of online multi-modal data fusion by constructing an offline large-scale rich-label detection dataset. This model moves the process of integrating multiple modalities online, as seen in Grounding DINO, to the data preparation phase, significantly reducing training costs. The authors of LAE-DINO explicitly state that their model is an improvement upon Grounding DINO. LAE-DINO’s DVC module dynamically selects positive and negative vocabularies for each training batch, addressing the issue of large-scale vocabulary sets. The VisGT module enhances text features by introducing "scene features." These modifications enable LAE-DINO to be trained using only four A100 GPUs, with pretraining requiring approximately 180K steps and taking about 48 GPU hours. Of course, this does not account for the cost of generating the pretraining dataset using a pre-trained model.

In summary, the core idea of best models in the field of open vocabulary and open set detection is to interact language and image features during training as much as possible, ensuring that the features used for bounding box regression and label classification are as rich in semantic and image information as possible. This contrastive learning approach is the source of zero-shot transfer capabilities. YOLO11-Earth explores a multi-modal method that does not involve contrastive learning training but simply integrates text features into the training process. Although our model does not achieve the zero-shot detection capability of models trained with contrastive learning on open sets, it factually endows the YOLO11 model, which should have no generalization capability, with zero-shot transfer capabilities on unseen datasets while retaining the fast fine-tuning capabilities of open set detectors (mainly compared to YOLO-Worldv2 and LAE-DINO). At the same time, the largest YOLO11x-Earth model only requires approximately 60 hours of pretraining on a single NVIDIA RTX 4090 GPU, significantly reducing pretraining costs. The intuitive training cost comparison of the four models is shown in Table 8. The V100 GPU is assumed to have 32GB of memory by default, and the A100 GPU is assumed to have 80GB of memory by default.

6.2 Text Embedding as Bias

To explain the principle behind YOLO11-Earth’s ability to achieve open vocabulary detection using only the text content of labels, we first consider the encodable label mapping brought by the WorldDetect. This detection head that receives text features ensures that the classification of bounding boxes is within the given label range and can perform a certain "affine transformation" on the classification results based on new text

Table 8: Comparison of training costs for different open vocabulary detection models.

Model	GPU Model (Count)	Max Memory	Cost (per hour, USD\$, 5/13/2025)
Grounding DINO	NVIDIA A100 (64)	5120GB	59.35
YOLO-World	NVIDIA V100 (32)	1024GB	8.35
LAE-DINO	NVIDIA A100 (4)	320GB	3.71
YOLO11-Earth	NVIDIA RTX4090 (1)	24GB	0.29

feature inputs, namely translation and rotation. Relying on this detection head, YOLO11-Earth inherits the "prompt first, then detect" strategy of the YOLO-World architecture.

Secondly, in the forward process of YOLO11-Earth, all label text information is converted into text features by the CLIP text encoder and added to the input of each C2FAttn module and the WorldDetect detection head during forward propagation. This "addition" operation is essentially an affine transformation of the input feature vectors of each layer. Constructing a simple mapping can help us understand this process. Let the input data be denoted as a vector x , $x \in \mathbb{R}^n$, and the forward process of each layer in a multilayer perceptron can be represented as $f_i = W_i x + b_i$, where $i \in [1, m]$, m is the number of layers in the neural network, and the activation function is denoted as σ . The output is y , and the mapping that the entire neural network attempts to fit is denoted as g . Assuming that all layers except the last one have activation functions, the output of a four-layer network can be written as:

$$y = g(x) = W_1 \cdot \sigma(W_2 \cdot \sigma(W_3 \cdot \sigma(W_4 x + b_4) + b_3) + b_2) + b_1 \quad (4)$$

In essence, a deep learning model is a complex non-linear mapping composed of a series of linearly combined parameter layers, and the parallel operations within each layer can be understood as operations on the blocks of the matrix $Wx + b$. Assuming the first layer represents the output of the model's backbone network (without text features t), then the operation of adding text features in the neck and detection head parts of YOLO11-Earth can rewrite the mapping as:

$$y = g(x) = W_1 \cdot \sigma(W_2 \cdot \sigma(W_3 \cdot \sigma(W_4 x + b_4 + t) + b_3 + t) + b_2 + t) + b_1 \quad (5)$$

Intuitively, the text feature t added at each step causes a fixed offset in the output of the previous layer. We can visualize the distributions of y obtained before and after adding t to test the above inference. Figure 9

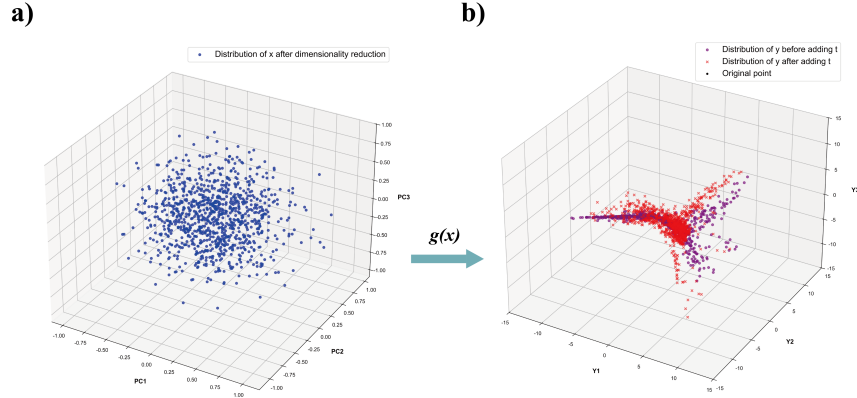


Figure 10: Visualization of the distribution shift caused by text features. (a) Original 6D input data reduced to 3D using PCA. (b) Output distribution before (purple) and after (red) adding text features, showing clear separation in the distribution spaces.

shows the change in the distribution of y before and after adding the constant feature t . The 6-dimensional input variable x is reduced to 3 dimensions using PCA for visualization purposes. A set of 1000 points randomly sampled from a 6-dimensional Gaussian distribution, with each dimension normalized to $[-1, 1]$, is used as input. A random 6-dimensional vector t is generated to mimic the text feature vector, and the distribution of y is obtained using the aforementioned formula. The weights and biases of the first three layers are set to ensure that the dimension of x remains unchanged (6 dimensions), and ReLU is used as the activation function after the third layer.

As can be observed, the distribution of y obtained using only image features (purple points in Figure 9b) exhibits a discernible difference from the distribution of y after incorporating the text feature t (red points in Figure 9b). This discrepancy is evident in both the mean and variance of the distributions. Consequently, when these distinct distributions are subjected to classification via a fully connected layer, the resulting classifications will naturally diverge.

This visualization elucidates why YOLO11-Earth, despite the absence of contrastive learning during training to align text and image features, still acquires a rudimentary capability for open-set detection. This capability stems from the affine transformation (vector addition) imposed on the intermediate output of each layer by the text features. This operation sacrifices the fitting of the original image feature distribution to some extent but enhances the alignment with the text feature distribution. The reason YOLO11-Earth’s zero-shot transfer capability falls short of models trained with contrastive learning alignment is that the text features in YOLO11-Earth remain constant throughout the forward pass and are merely added to the image features. This offset is insufficiently strong to achieve robust alignment.

In conclusion, YOLO11-Earth’s open vocabulary detection capability, while not on par with models trained using contrastive learning, is nonetheless a significant achievement given its minimal training cost. This capability is rooted in the affine transformation induced by the text features, which, despite its simplicity, enables the model to generalize to unseen datasets. This finding underscores the potential for lightweight models to achieve open vocabulary detection with significantly reduced computational resources, paving the way for more accessible and efficient remote sensing applications.

6.3 Limitations

Despite the significant improvements offered by YOLO11-Earth in the context of open vocabulary detection for remote sensing imagery, several limitations persist. One notable limitation is its terrible zero-shot transfer capability when compared to models trained with contrastive learning. This is primarily attributed to YOLO11-Earth’s reliance on closed-set training data, which inherently restricts its ability to generalize across diverse datasets. The model’s performance is further constrained by the absence of updated text features and the lack of specialized loss functions designed to optimize text feature learning. These factors collectively result in YOLO11-Earth underperforming in zero-shot scenarios where models are expected to recognize novel categories without additional training.

Moreover, the model encounters challenges when deployed across different remote sensing datasets due to the inherent variability in sensor types and imaging conditions. This is exacerbated by the specific characteristics of remote sensing data, such as high dimensionality and complex background noise, which can lead to suboptimal performance if not adequately addressed. The model’s architecture, while efficient, may not fully capture the intricate patterns and nuances present in remote sensing imagery, thereby limiting its detection accuracy and reliability.

7 Conclusion

This technical report has introduced YOLO11-Earth, a lightweight open vocabulary detection model designed for remote sensing imagery. Built upon the foundations of YOLO11 and YOLOv8-Worldv2, YOLO11-Earth leverages model pruning and innovative VL-PAN design to achieve real-time open vocabulary detection capabilities on conventional datasets and consumer-grade GPUs. Extensive experiments across various benchmarks have demonstrated YOLO11-Earth’s superior performance in terms of detection accuracy, computational efficiency, and cross-dataset generalization. These findings underscore the model’s potential to serve as a versatile and cost-effective solution for a wide array of remote sensing applications.

However, the report also highlights several avenues for future improvement. Enhancing YOLO11-Earth’s zero-shot transfer learning capabilities represents a key area for development. This could be achieved by refining the model’s text feature encoding mechanisms and incorporating more sophisticated cross-modal alignment techniques. Additionally, expanding the diversity and scale of training datasets would better equip the model to handle the complexities and variabilities of real-world remote sensing data. Further exploration into advanced model architectures and training strategies, such as incorporating attention mechanisms and employing more efficient feature extraction methods, could also significantly boost YOLO11-Earth’s performance. The ultimate goal is to create a more powerful and adaptable open vocabulary detection model that can reliably and accurately detect objects across various remote sensing datasets and scenarios.

In summary, while YOLO11-Earth has made significant strides in the field of lightweight remote sensing open vocabulary detection, ongoing research and development efforts are necessary to overcome its limitations and unlock its full potential. By addressing the identified challenges, future open-vocabulary detectors can be expected to deliver even more impressive results and further advance the capabilities of remote sensing object detection technologies, with low cost and high performance.

References

- [1] A. Zareian, K. D. Rosa, D. H. Hu, and S.-F. Chang, "Open-vocabulary object detection using captions," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 14 393–14 402.
- [2] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boulton, "Toward open set recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 35, no. 7, pp. 1757–1772, 2012.
- [3] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 580–587.
- [4] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*. Springer, 2016, pp. 21–37.
- [5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified real-time object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 779–788.
- [6] F. C. Akyon, S. O. Altinuc, and A. Temizel, "Slicing aided hyper inference and fine-tuning for small object detection," in *2022 IEEE international conference on image processing (ICIP)*. IEEE, 2022, pp. 966–970.
- [7] X. Wu, D. Hong, and J. Chanussot, "Uiu-net: U-net in u-net for infrared small object detection," *IEEE Transactions on Image Processing*, vol. 32, pp. 364–376, 2022.
- [8] C. Yang, Z. Huang, and N. Wang, "Querydet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proceedings of the IEEE/CVF Conference on computer vision and pattern recognition*, 2022, pp. 13 668–13 677.
- [9] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "Yolo-world: Real-time open-vocabulary object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 901–16 911.
- [10] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su *et al.*, "Grounding dino: Marrying dino with grounded pre-training for open-set object detection," in *European Conference on Computer Vision*. Springer, 2024, pp. 38–55.
- [11] T. Ren, Q. Jiang, S. Liu, Z. Zeng, W. Liu, H. Gao, H. Huang, Z. Ma, X. Jiang, Y. Chen *et al.*, "Grounding dino 1.5: Advance the "edge" of open-set object detection," *arXiv preprint arXiv:2405.10300*, 2024.
- [12] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1440–1448.
- [13] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *IEEE transactions on pattern analysis and machine intelligence*, vol. 39, no. 6, pp. 1137–1149, 2016.
- [14] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2961–2969.
- [15] J. Redmon and A. Farhadi, "Yolo9000: better, faster, stronger," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 7263–7271.
- [16] —, "Yolov3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.
- [17] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "Yolov4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.
- [18] C. Li, L. Li, H. Jiang, K. Weng, Y. Geng, L. Li, Z. Ke, Q. Li, M. Cheng, W. Nie *et al.*, "Yolov6: A single-stage object detection framework for industrial applications," *arXiv preprint arXiv:2209.02976*, 2022.

- [19] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, “Yolov7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 7464–7475.
- [20] Y. Tian, Q. Ye, and D. Doermann, “Yolov12: Attention-centric real-time object detectors,” *arXiv preprint arXiv:2502.12524*, 2025.
- [21] G. Jocher, A. Stoken, J. Borovec, L. Changyu, A. Hogan, L. Diaconu, J. Poznanski, L. Yu, P. Rai, R. Ferriday *et al.*, “ultralytics/yolov5: v3. 0,” *Zenodo*, 2020.
- [22] M. Sohan, T. Sai Ram, and C. V. Rami Reddy, “A review on yolov8 and its advancements,” in *International Conference on Data Intelligence and Cognitive Informatics*. Springer, 2024, pp. 529–545.
- [23] C.-Y. Wang, I.-H. Yeh, and H.-Y. Mark Liao, “Yolov9: Learning what you want to learn using programmable gradient information,” in *European conference on computer vision*. Springer, 2024, pp. 1–21.
- [24] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han *et al.*, “Yolov10: Real-time end-to-end object detection,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 107 984–108 011, 2024.
- [25] R. Khanam and M. Hussain, “Yolov11: An overview of the key architectural enhancements,” *arXiv preprint arXiv:2410.17725*, 2024.
- [26] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, “End-to-end object detection with transformers,” in *European conference on computer vision*. Springer, 2020, pp. 213–229.
- [27] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, “Dino: Detr with improved denoising anchor boxes for end-to-end object detection,” *arXiv preprint arXiv:2203.03605*, 2022.
- [28] F. Li, H. Zhang, H. Xu, S. Liu, L. Zhang, L. M. Ni, and H.-Y. Shum, “Mask dino: Towards a unified transformer-based framework for object detection and segmentation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2023, pp. 3041–3050.
- [29] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, “Detrs beat yolos on real-time object detection,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2024, pp. 16 965–16 974.
- [30] W. Lv, Y. Zhao, Q. Chang, K. Huang, G. Wang, and Y. Liu, “Rt-detr2: Improved baseline with bag-of-freebies for real-time detection transformer,” *arXiv preprint arXiv:2407.17140*, 2024.
- [31] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.
- [32] X. Gu, T.-Y. Lin, W. Kuo, and Y. Cui, “Open-vocabulary object detection via vision and language knowledge distillation,” *arXiv preprint arXiv:2104.13921*, 2021.
- [33] L. H. Li, P. Zhang, H. Zhang, J. Yang, C. Li, Y. Zhong, L. Wang, L. Yuan, L. Zhang, J.-N. Hwang *et al.*, “Grounded language-image pre-training,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 10 965–10 975.
- [34] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, “Swin transformer: Hierarchical vision transformer using shifted windows,” in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 10 012–10 022.
- [35] H. Zhang, P. Zhang, X. Hu, Y.-C. Chen, L. Li, X. Dai, L. Wang, L. Yuan, J.-N. Hwang, and J. Gao, “Glipv2: Unifying localization and vision-language understanding,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 36 067–36 080, 2022.
- [36] L. Liu, J. Feng, H. Chen, A. Wang, L. Song, J. Han, and G. Ding, “Yolo-uniow: Efficient universal open-world object detection,” *arXiv preprint arXiv:2412.20645*, 2024.
- [37] Y. Sun, J. Wang, Y. You, Z. Yu, S. Bian, E. Wang, and W. Wu, “Yolo-e: a lightweight object detection algorithm for military targets,” *Signal, Image and Video Processing*, vol. 19, no. 3, p. 241, 2025.
- [38] J. Pan, Y. Liu, Y. Fu, M. Ma, J. Li, D. P. Paudel, L. Van Gool, and X. Huang, “Locate anything on earth: Advancing open-vocabulary object detection for remote sensing community,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 6, 2025, pp. 6281–6289.
- [39] Z. Huang, Y. Feng, S. Yang, Z. Liu, Q. Liu, and Y. Wang, “Openrsd: Towards open-prompts for object detection in remote sensing images,” *arXiv preprint arXiv:2503.06146*, 2025.

- [40] D. Lam, R. Kuzma, K. McGee, S. Dooley, M. Laielli, M. Klaric, Y. Bulatov, and B. McCord, “xview: Objects in context in overhead imagery,” *arXiv preprint arXiv:1802.07856*, 2018.
- [41] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, “Object detection in optical remote sensing images: A survey and a new benchmark,” *ISPRS journal of photogrammetry and remote sensing*, vol. 159, pp. 296–307, 2020.
- [42] ISPRS, “Benchmark on semantic labeling,” Online: <https://www.isprs.org/education/benchmarks/UrbanSemLab/Default.aspx>, 2018, accessed: 2025-05-13.
- [43] D. Wang, J. Zhang, M. Xu, L. Liu, D. Wang, E. Gao, C. Han, H. Guo, B. Du, D. Tao *et al.*, “Mtp: Advancing remote sensing foundation model via multi-task pretraining,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024.
- [44] K. Ayush, B. UzKent, C. Meng, K. Tanmay, M. Burke, D. Lobell, and S. Ermon, “Geography-aware self-supervised learning,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 181–10 190.
- [45] U. Mall, B. Hariharan, and K. Bala, “Change-aware sampling and contrastive learning for satellite images,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 5261–5270.
- [46] C. Tao, J. Qi, G. Zhang, Q. Zhu, W. Lu, and H. Li, “Tov: The original vision model for optical remote sensing image understanding via self-supervised learning,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 16, pp. 4916–4930, 2023.
- [47] C. J. Reed, R. Gupta, S. Li, S. Brockman, C. Funk, B. Clipp, K. Keutzer, S. Candido, M. Uyttendaele, and T. Darrell, “Scale-mae: A scale-aware masked autoencoder for multiscale geospatial representation learning,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 4088–4099.
- [48] F. Bastani, P. Wolters, R. Gupta, J. Ferdinando, and A. Kembhavi, “Satlaspretrain: A large-scale dataset for remote sensing image understanding,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 16 772–16 782.
- [49] X. Sun, P. Wang, W. Lu, Z. Zhu, X. Lu, Q. He, J. Li, X. Rong, Z. Yang, H. Chang *et al.*, “Ringmo: A remote sensing foundation model with masked image modeling,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–22, 2022.
- [50] X. Guo, J. Lao, B. Dang, Y. Zhang, L. Yu, L. Ru, L. Zhong, Z. Huang, K. Wu, D. Hu *et al.*, “Skysense: A multi-modal remote sensing foundation model towards universal interpretation for earth observation imagery,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 672–27 683.